5 The internet and cyber security

1 a

Internet	**World wide web (www)**
Users can send and receive emails	It is a collection of multimedia web pages and other information on websites
Allows online chatting (via text, audio and video)	Web pages are written using hypertext mark-up language (html)
Makes use of transmission protocols (TCP) and internet protocols (IP)	Uniform resource locators (URLs) are used to specify the locations of web pages
It is a worldwide collection of interconnected networks and devices	Web resources are accessed by web browsers
	The www uses the internet to access information from web servers
	Makes use of the http(s) protocol

b i https

ii www

iii .com

iv comp_science_sample

c Three from:

- They have a home page.
- They can store a user's favourite websites/web pages (used as a bookmark).
- They keep a history of websites visited by the user (user history).
- They have the ability to allow the user to navigate forwards and backwards through websites/web pages already opened; many webpages can be open at the same time (navigation).
- They make use of hyperlinks that allow navigation between websites and web pages; links are shown in one of two ways:
  - either the website is underlined and requires <ctrl> + <click> to follow the link
  - or a small pointed finger shows under the name of the website link to highlight it
- They make use of cookies.
- Data is stored as a cache.
- They translate HTML and display the result of the translation on the user's device.
- They make use of JavaScript.
- They use an address bar.

2 order: 2, 4, 5, 1, 6, 3

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3 a
- a cookie that is stored temporarily on a computer
- cookie is deleted when the browser is closed or the website session terminated

b
- a cookie that is stored on a user's HDD/SSD
- a cookie that is only deleted when its expiry date is reached or it is deleted by the user

c Three from:
- They allow a website to remember a user's details.
- They allow a website to recognise a user every time they visit the website.
- They can save a user's items in a virtual shopping basket.
- They can track a user's internet habits and their browsing history or favourites (bookmarks).
- They can target users with advertising which matches previous buying/surfing habits.
- They can store user preferences.
- They allow different languages to be used on the webpage automatically as soon as a user logs onto the website.
- They allow progress in online games/quizzes to be stored.

4 a

Blockchaining statement	**True (✓)**	**False (✓)**
All digital currency systems use blockchaining		✓
Blockchaining uses a decentralised database	✓	
The last block in a blockchain is known as the 'genesis block'		✓
New hash values are only generated when data in a block is altered		✓
Tampering of data in a block by a hacker would cause the hash value to change	✓	
Blocks are 'policed' by network users called 'miners'	✓	

b In any order:
- data (for example, name of sender/recipient)
- hash value (a unique 'fingerprint')
- previous hash value (pointing back to previous block in the chain)

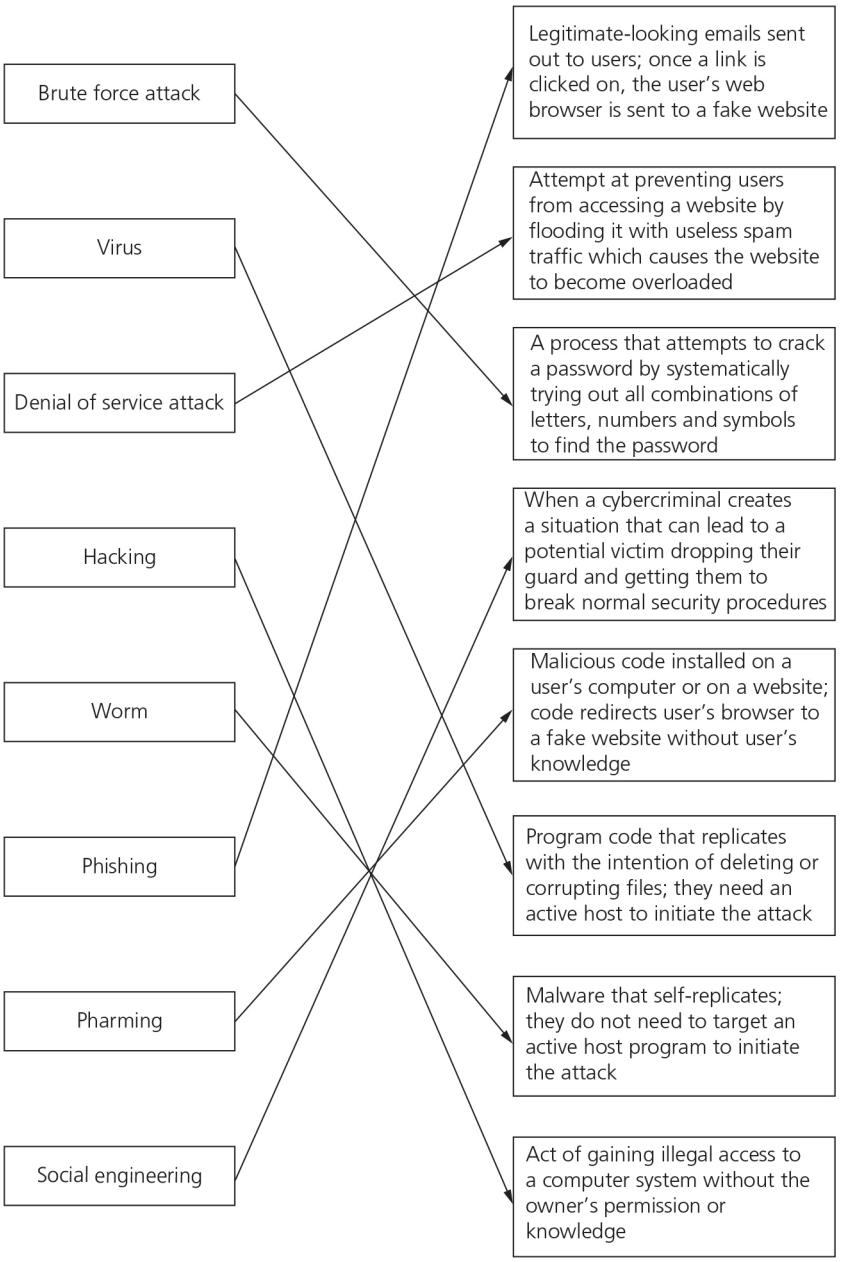
c i 0000 A4BF 6AB1 34EE FF12 AA00

ii Any changes to block '4' will cause the hash value (FF12) to be changed. This means that blocks '5' and '6' and beyond will now be invalid because the chain was broken between '4' and '5' (previous hash value FF12 is no longer valid).

d
- Cryptocurrency uses cryptography to track transactions.
- Digital currency uses a central banking system; whereas cryptocurrency is decentralised.
- Cryptocurrency uses block chaining.
- Cryptocurrency transactions are publicly available and therefore all transactions can be tracked.

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5



6 a i Trojan horse: a malicious program often disguised as legitimate-looking software; it replaces all or part of the genuine software with the intent of carrying out some harm to the user's system.

ii Adware: software that floods the user's computer with unwanted advertising; usually in the form of 'pop ups' but can frequently appear in the browser's address window redirecting the browser to another website that contains promotional adverts (or sometimes adverts that create a security breach).

iii Ransomware: programs that encrypt the data on a user's computer; a decryption key is sent back to the user once they pay a sum of money (the ransom); often sent via a Trojan horse or by social engineering.

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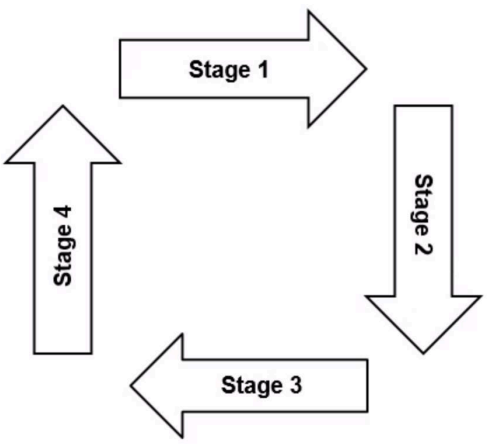
b i This occurs when a cybercriminal creates a social situation that can lead to a potential victim 'dropping their guard'. It involves manipulation of people into breaking their normal security protocols by:

- instant messaging
- scareware
- email/phishing scams
- baiting
- bogus phone calls

ii

Threat used by cybercriminals	**How the threat is used (include examples)**
Instant messaging	• malicious links are embedded into instant messages • for example, an important software upgrade • relies on the user's curiosity
Scareware	• this is often done using a pop-up message that claims that the user's computer is infected with a virus • the user is told they need to download the fake antivirus immediately • relies on the user's fear of viruses and other malware
Baiting	• the cybercriminal leaves a malware-infected memory stick somewhere where it can be found: • the finder picks up the memory stick and plugs it into their computer (just to see who it belongs to) • the finder unwittingly downloads malicious malware • relies on the user's curiosity
Phone calls	• a so-called IT professional calls the user on their mobile claiming their device has been compromised in some way • the user is advised to download some special software that allows the cybercriminal to take over the user's device … • … giving them access to personal information • relies on fear of the 'unknown' risks

c



Stage 1 – the victims are identified; information about victim gathered and method of malware attack decided

Stage 2 – at this stage the victim is being targeted (either through email, phone call, Trojan horse and so on; it all depends on who the victim is)

Stage 3 – the attack on the victim is now executed allowing the cybercriminal to obtain the information or to cause the disruption decided on at Stage 1

Stage 4 – when the cybercriminal has decided they have what they wanted they try to remove all traces of the malware to cover their tracks

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7 a i It is software that detects and removes spyware programs installed illegally on a user's computer system.

ii Three from:

- It detects and removes spyware already installed on a device.
- It prevents a user from downloading spyware.
- It encrypts files to make the data more secure in case it is 'spied' on.
- It encrypts keyboard strokes to help remove the risk posed by the keylogging aspects of some spyware.
- It blocks access to a user's webcam and microphone (the software stops the spyware from taking over the control of a user's webcam and microphone which can be used to collect information without the user's knowledge).
- It scans for signs that the user's personal information has been stolen and warns the user if this has happened.

b

Biometric technique	**Benefits**	**Drawbacks**
Fingerprint or thumbprint scans	• is one of the most developed biometric techniques • has high accuracy • is very easy to use • has relatively small storage requirements for the biometric data created	• is very intrusive for some people because it is still related to criminal identification • can make mistakes if the skin is dirty or damaged (e.g. cuts)
Retina scans	• has very high accuracy • no known way to replicate a person's retina	• is very intrusive • can be relatively slow to verify the retina scan with stored scans • is very expensive to install and set up
Voice recognition	• non-intrusive method • verification takes less than 5 seconds • relatively inexpensive technology	• a person's voice can be easily recorded and used for unauthorised access • low accuracy • an illness such as a cold can change a person's voice, making absolute identification difficult or impossible

c

- A microphone picks up verbal commands.
- Certain words, such as 'navigate', 'media', 'open windows', etc, often need to be used.
- The processor processes words by comparing them to those stored on the database.
- Commands are sent to a device to operate satnav, media or window mechanism.
- Signals are sent to actuators to operate motors to open/close windows, and so on.

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8 **a** order: 4, 5, 2, 6, 3, 1

b They contain 'patches' that contain software improvements, for example, removing glitches or adding new features; updates may contain security fixes, such as improved anti-malware routines.

9 Four from:

- email address gives clues, for example, no company will use an email such as '…@gmail.com'
- email address should contain the company name
- tone of email is a clear indication of a scam
- not only spelling mistakes, but the language used and the way the target victim is addressed, for example 'Dear User'
- mis-spelling in a subtle way (i.e. typo squatting)
- suspicious links; the destination address should match the rest of the email
- some links (such as, How To Contact Us) don't work
- look at the domain; for example, large companies would use .com rather than .co or .org

10 **a** a type of protocol that:

- allows data to be sent and received securely over a network
- encrypts data
- is active when the connection protocol is https or if there is a small green padlock by the website address

b order: 2, 4, 1, 5, 3

c Three from:

- online banking and financial transactions
- online shopping/commerce
- when sending out software to a restricted list of users
- sending/receiving emails
- using cloud storage facilities
- using VoIP
- when using instant messaging
- when using social networking sites

11 **a**

- They can filter information in and out of a computer.
- They are the primary defence of any computer system to prevent it from hacking, malware, phishing and pharming.
- They can be hardware or software.

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